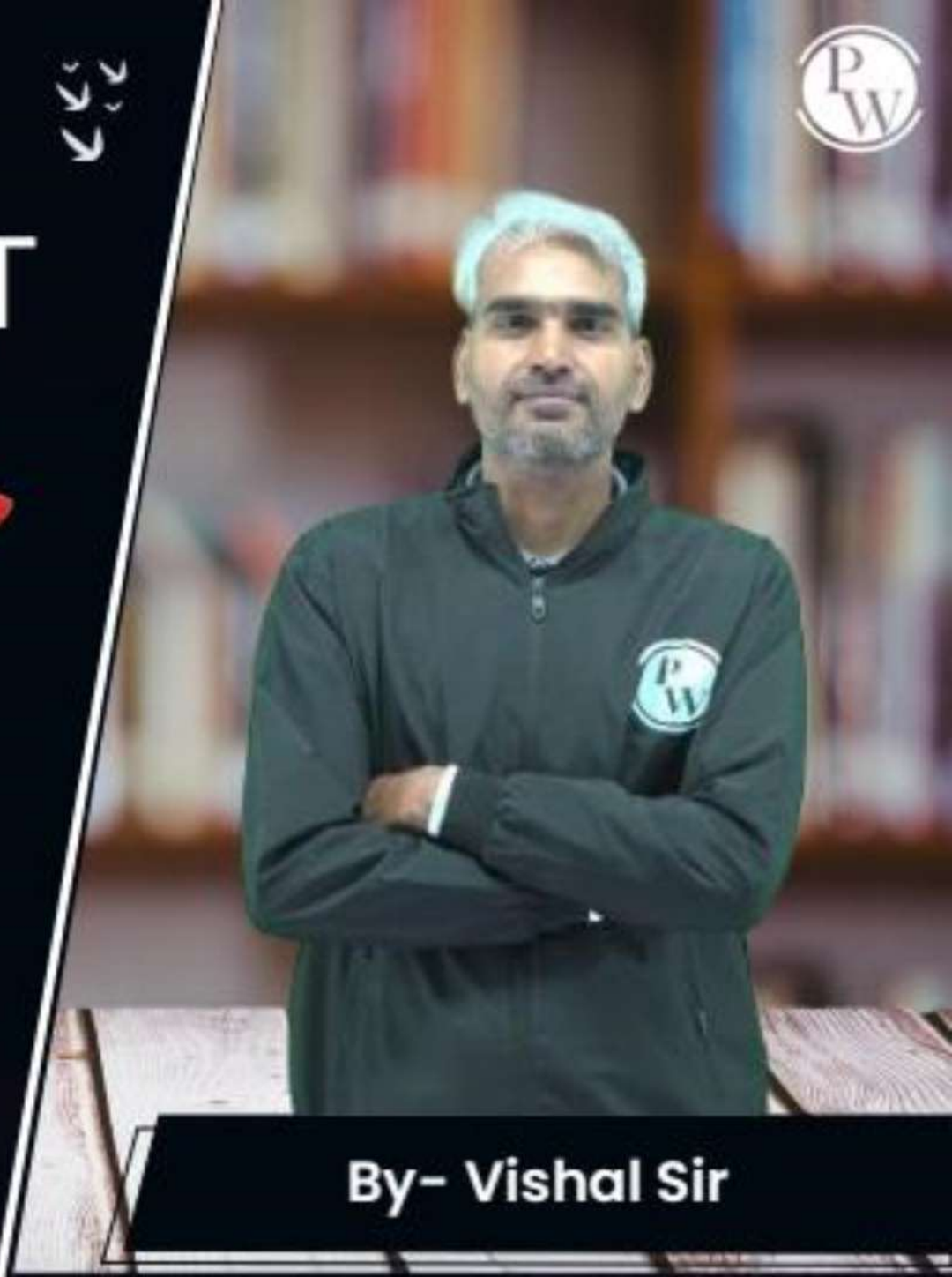
Computer Science & IT



Compiler Design

Syntax Directed Translation

Lecture No. 01



By- Vishal Sir

Recap of Previous Lecture



Topic Operator Precedence Parsing

Topic Evaluation of expression based on grammar

Slide

Topics to be Covered



Topic	SDD and SDT
Topic	Types of attributes
Topic	S-attributed and L-attributed SDT
Topic	Questions based on S-attributed SDT



Topic : Syntax Directed Definition

- It is a CFG along with attributes and semantic rules (translation rules).
- Attributes are associated with the grammar symbols. → i.e., Non-terminals
- Semantic rules are associated with the productions of the grammar.

eg:

CFG	Semantic Rules
$E \rightarrow E_1 + T$	$\{E.val = E_1.val + T.val\}$
$E \rightarrow T$	$\{E.val = T.val\}$
$T \rightarrow T_1 * F$	$\{T.val = T_1.val * F.val\}$
$T \rightarrow F$	$\{T.val = F.val\}$
$F \rightarrow id$	$\{F.val = id.lexval\}$

Value is an attribute

Syntax Directed Definition.



Topic : Syntax Directed Translation

- In syntax directed translation we parse the input token stream, build the parse tree, and traverse the tree as needed (based on dependency induced by semantic rules) to evaluate the semantic rules at the parse tree nodes.
- Evaluation of these semantic rules:
  - may generate intermediate codes
  - may put information into the symbol table
  - may perform type checking
  - may issue error messages
  - may perform some other activities

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Topic : Semantic actions

written in Programming language

- In syntax directed translation, Semantic actions are fragment of code which are embedded within the production body.

eg ① $S \rightarrow x\,x\,W$ { Printf("1"); }

② $S \rightarrow y$ { Printf("2"); }

③ $W \rightarrow S\,z$ { Printf("3"); }

fragments of Code

∴ these are known as Semantic actions

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Topic : Attribute

An attribute may correspond to almost any thing.

– a string, a number, data type, a memory location etc..

- There are two types of attributes

1) Synthesized attributes { depends on Children }

2) Inherited attributes { depends on parent and/or siblings }

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Topic : Synthesized attributes

- The value of an attribute at a node of parse tree can be obtained from the values of attributes at its child nodes. , then that attribute is called Synthesized attribute

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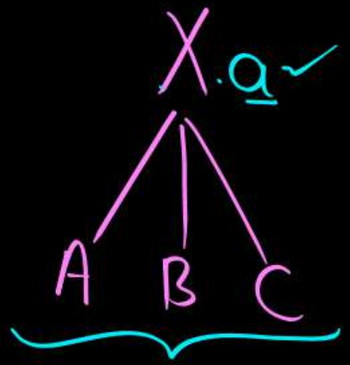
Topic : Inherited attributes

- The value of attribute at a node of parse tree can be obtained from the values of attributes at its parent node and/or its sibling nodes., then it is called inherited attribute

eg:-

$X \longrightarrow ABC \{X.a = A.b + B.b + C.a\}$

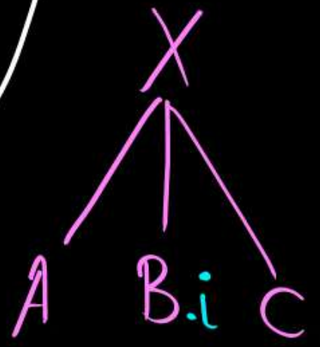
Synthesized Attribute



A, B and C are children of X.

$X \rightarrow ABC \{B.i = X.i + A.b\}$

inherited attribute



Wrt. B

X: is parent of B

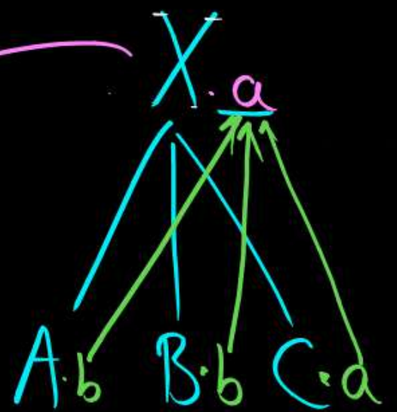
A: left sibling of B

C: Right sibling of B

Note:- Value of synthesized attribute at a node Can be computed during bottom-up parsing

$X \longrightarrow ABC$

$X.a = A.b + B.b + C.a$



green colour arrow represent dependency graph

it is acyclic

While reducing ABC into X during bottom-up parsing We have already seen A, B and C

∴ We will be able to compute the value of attribute of non-terminal X if the value of that attribute depends on attributes of A, B and C



Topic : Types of SDT

There are two types of SDT

(1) S-attributed Definition / S-attributed SDT { only synthesized attributes are allowed }

(2) L-attributed Definition / L-attributed SDT

{ Both synthesized and inherited attributes are allowed, but with some restrictions }

Slide



Topic : S-attributed SDT

- If an SDT uses only synthesized attributes, then it is called as S-attributed SDT.
- S-attributed SDTs can be evaluated in bottom-up manner, as the values of the parent nodes depend only upon the values of the child nodes.
- Semantic rules are placed at rightmost end of production, therefore S-attributed SDTs are also known as postfix SDTs.

eg: $X \longrightarrow ABC\{\ \}$ Semantic Rule



Topic : L-attributed SDT

- L-attributed SDT can use both synthesized attributes as well as inherited attributes with a restriction that inherited attribute can inherit values only from its left siblings and/or its parent (only inherited attributes of the parent node can be used. We can not use the synthesized attributes of parent node), then it is called as L-attributed SDT.
- Attributes in L-attributed SDTs are evaluated in depth-first manner from left-to-right.
- Semantic rules can be placed anywhere in RHS.

Eg let Production is $X \rightarrow ABC$

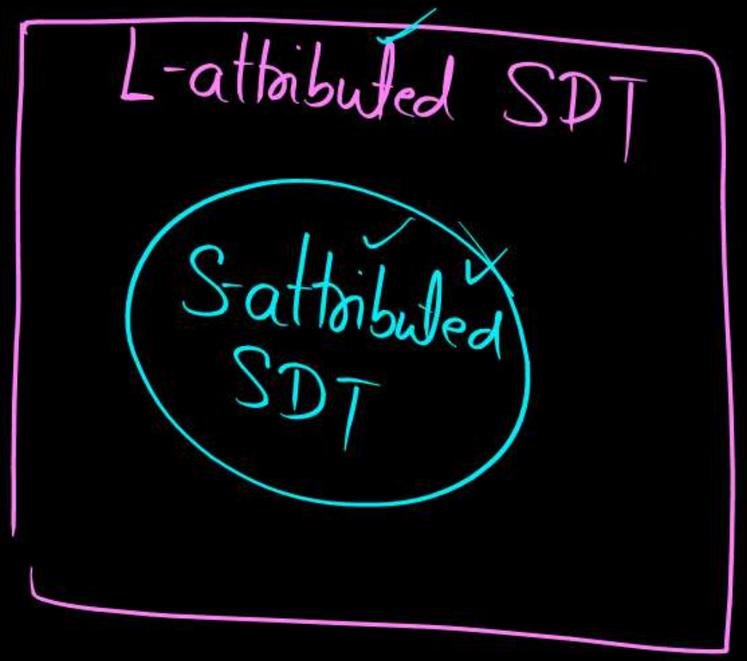
$X \rightarrow ABC\{__\}$

$X \rightarrow AB\{__\}C$

$X \rightarrow A\{__\}BC$

Semantic Rules

Note:- Every S-attributed definition is L-attributed, but every L-attributed definition need not be S-attributed





Topic : Rule for L-attributed SDT

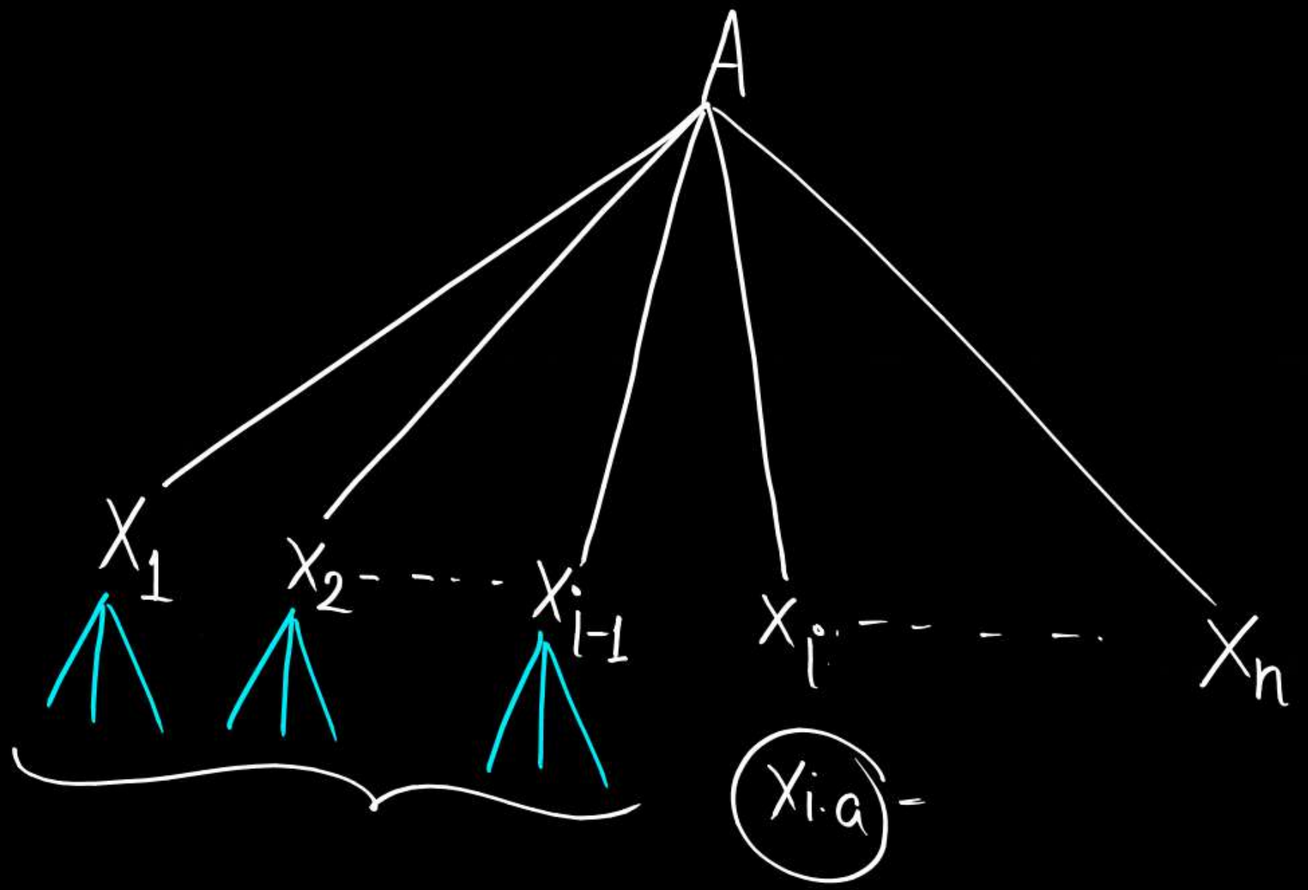
- For a production $A \rightarrow X_1\ X_2\ X_3\ X_4 \ldots\ldots\ldots X_n$,

  Attribute of Symbol Xi can be computed using

  1. Inherited attributes associated with the parent node A.
  2. Either inherited or synthesized attributes associated with the occurrences of symbols $X_1\ X_2\ X_3\ X_4 \ldots\ldots\ldots$ Xi-1 located to the left of Xi.
  3. Inherited or synthesized attributes associated with this occurrence of Xi itself.

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$A \longrightarrow X_1\ X_2\ X_3 \cdots X_{i-1}\ X_i\ X_{i+1} \cdots X_n$



2 Min

Q1. Consider the productions A → PQ and A → XY. Each of the five non-terminals A, P, Q, X and Y has two attributes: s is a synthesized attribute, and i is an inherited attribute. Consider the following rules.

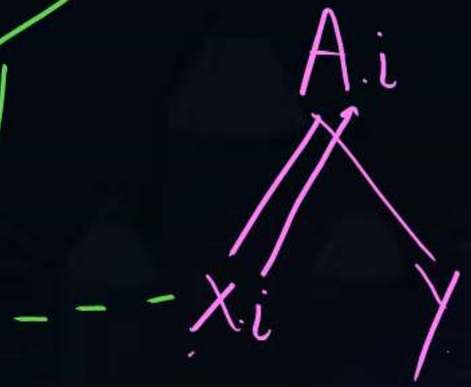
L-attributed

Rule 1: P.i = A.i + 2, Q.i = P.i + A.i, and A.s = P.s + Q.s

Not L-attributed

Rule 2: X.i = A.i + Y.s and Y.i = X.s + A.i

Which one of the following is TRUE ?



A. Both Rule 1 and Rule 2 are L-attributed.

B. Only Rule 1 is L-attributed

C. Only Rule 2 is L-attributed

D. Neither Rule 1 nor Rule 2 is L-attributed.

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Q2. Consider the following Production.

{ i is inherited attribute
& is Synthesized }

$A \rightarrow BC$

(0) $B.i = f(A.s)$ ✗

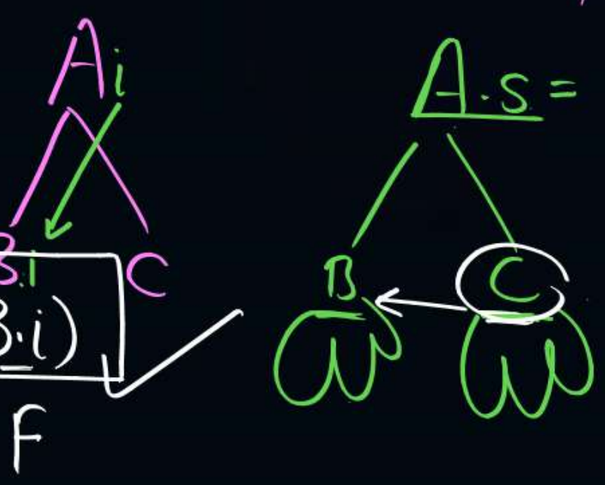
(I) $B.i = f(A.i)$ ✓

(II) $B.i = f(C.i)$ ✗

(III) $A.s = f(B.s)$ ✓

$C.i = f(B.i)$

T F



$A.s = f(B.s, B.i, C.s, C.i)$

Which of the above is violating L – attributed definition?

A. I only

B. II only

C. I and II

D. I, II and III

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Topic : Evaluation of S-attributed SDT

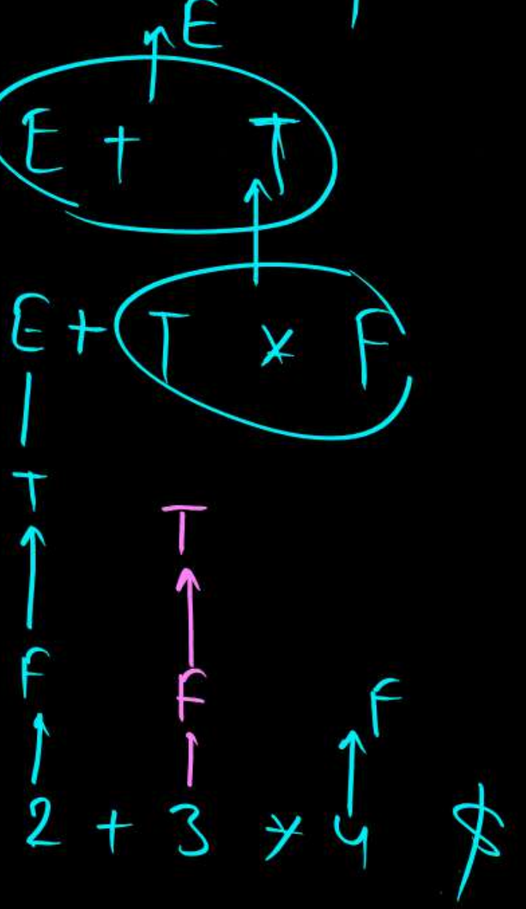
Q. Consider the Grammar with the following translation rule and E as the start symbol

$E \rightarrow E + T$	{E. Val = E.Val + T. Val}
$E \rightarrow T$	{E.Val = T.Val}
$T \rightarrow T*F$	{T. Value = T. value * F. Val}
$T \rightarrow F$	{T. Value = F. Val}
$F \rightarrow num$	{F. Val = num.Val}

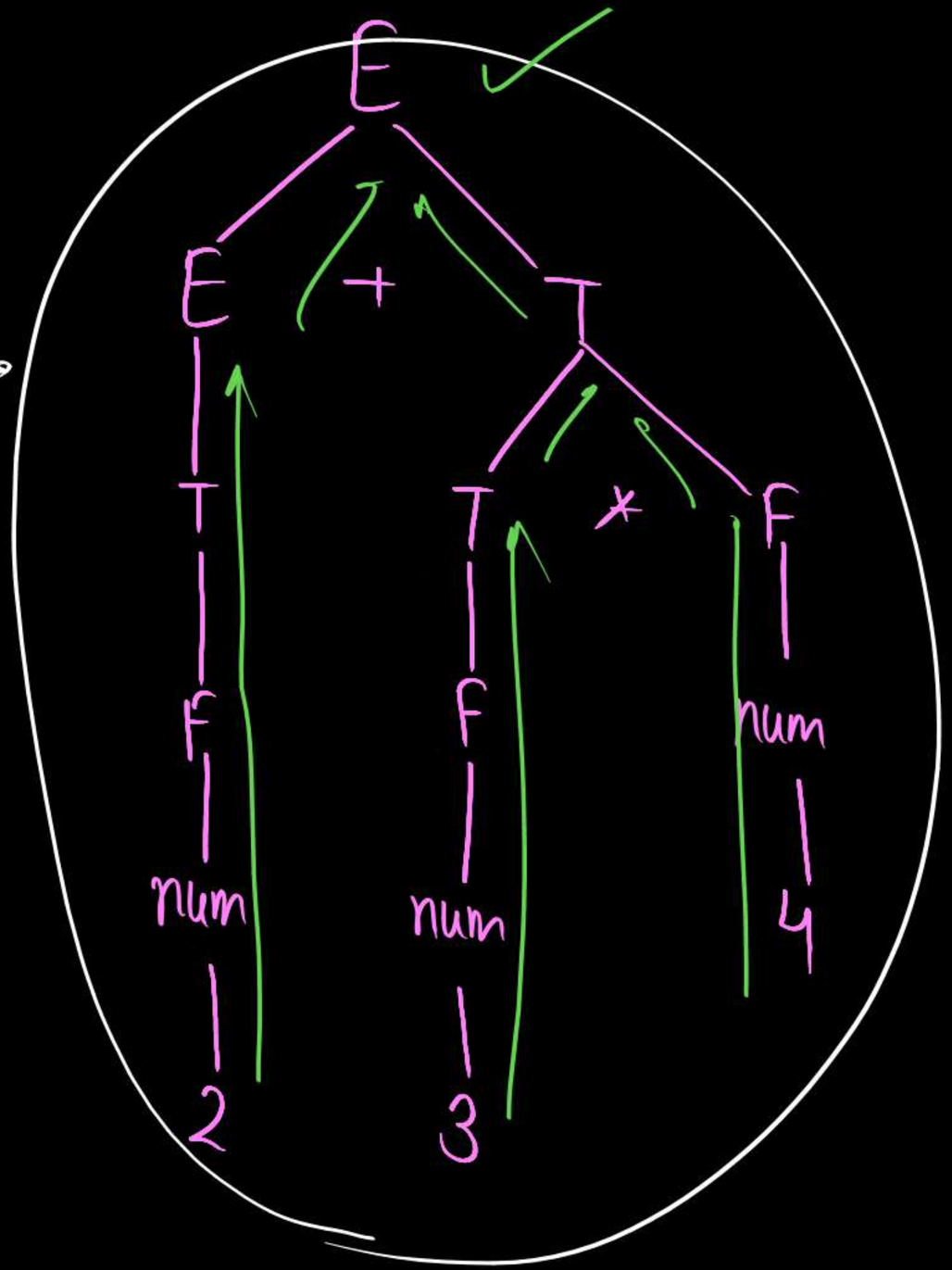
- What output will be produced if the input string 2+3*4 is given to the bottom-up parser.

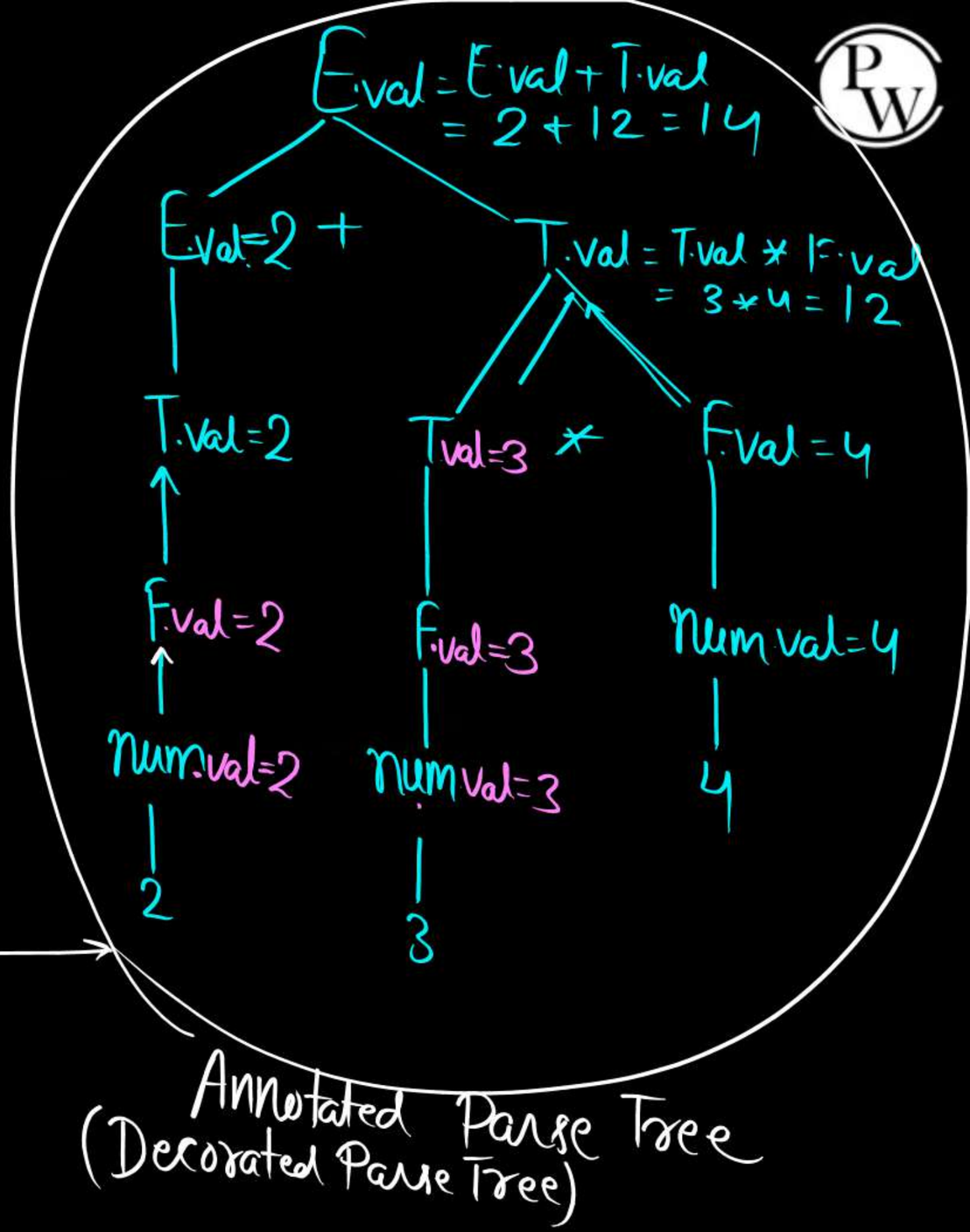
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* If SDT is given to bottom-up parser, then whenever reduce action is performed the semantic rule associated with that production will be evaluated



Parse Tree





Annotated Parse Tree
(Decorated Parse Tree)



Topic : Evaluation of S-attributed SDT

Precedence $\& > \#$ $\Longrightarrow$ $\# \equiv *$ $\Rightarrow$ $+ > *$

$\& \equiv +$

Q. Consider the Grammar with the following translation rule and E as the start symbol

$E \rightarrow E \# T$	{E. value = E. value * T. value}
$E \rightarrow T$	{ E. value = T. value}
$T \rightarrow T \& F$	{ T. value = T. value + F value}
$T \rightarrow F$	{ T. value = F. value}
$F \rightarrow num$	{F. value = num. value}

Compute E.value for the root of the parse tree for the expression

2 # 3 & 5 # 6 & 4 $\equiv$ $2 * (3+5) * (6+4) = 160$

$2 * 8 * 10$

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Topic : Evaluation of S-attributed SDT

Q. Consider the following SDT

$E \rightarrow E \# T$	{E. val = E. val * T. val}
$E \rightarrow T$	{E. val = T. val}
$T \rightarrow T \& F$	{ ? }
$T \rightarrow F$	{T. val = F. val}
$F \rightarrow num$	(F. val = num)

If the expression 8 # 12 & 4 # 16 & 12 # 4 & 2 is evaluated to 512 by the bottom-up parser, then which of the following is correct replacement for the ? .

A. T. val = T. val * F.val

B. T. Val = T. val + F val

C. T. val = T. val – F.val

D. None

$8 * 8 * 4 * 2 = 2^9 = 512$

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2 mins Summary

Topic SDD and SDT

Topic Types of SDT

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THANK - YOU

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